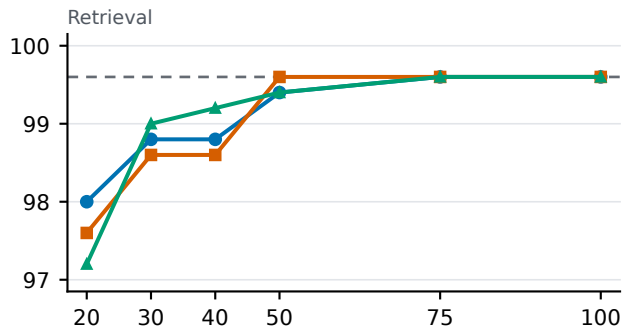


RULER 4K individual tasks (2/3)

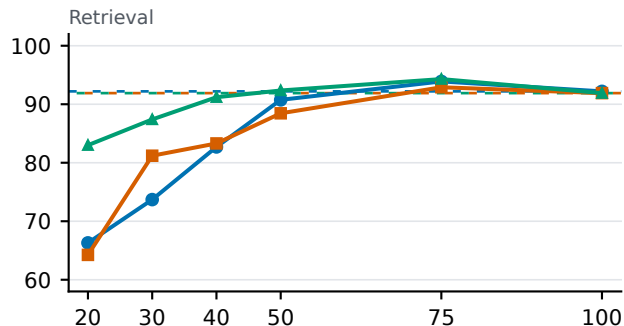
Graph-based selector FastKVzip KVzip

Dashed: full cache (method color; gray if shared). Y-axis range varies by panel.

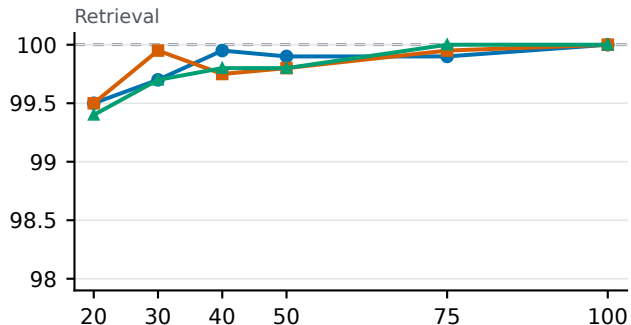
Multi-keys NIAH 3



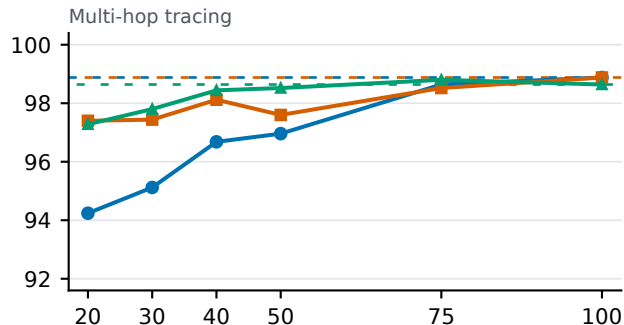
Multi-values NIAH



Multi-queries NIAH



Variable tracking



Retention ratio (%)

Each panel is one evaluated 4K configuration. All 13 RULER task configurations are included across this section.